Rigorous Formulation of the Post-Mitigated Subspace Filter: Theorems, Lemmas, and Stochastic Invariants

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Abstract

This document establishes the formalized, non-circular mathematical boundaries for the Eigenvector Cognitive Filter (ECF) and the Quicksand Oja++ online tracking algorithm. We present the core hypotheses, definitions, theorems, and hardware-aware stochastic invariants that govern tracking stability on the Stiefel manifold under high-entropy adversarial noise. By treating microarchitectural limitations as stochastic noise bounds rather than algebraic absolutes, and decoupling empirical and population spectra via Weyl's Perturbation Theorem, this framework provides a rigorous, localized foundation for stable, out-of-distribution subspace projection.

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1 Foundational Core Hypotheses & Framework Definitions

Hypothesis 1.1 (The Neuro-Spectral-Symbolic Equivalence Hypothesis)**.** Continuous neural vector fields $\mathcal{V} \subseteq \mathbb{R}^d$, linear spectral subspaces $\mathcal{W} \in \mathbb{G}(k, d)$, and discrete symbolic manifolds $\mathcal{M}$ can be mapped into a unified cognitive state space Ω without loss of semantic weights, provided all inter-granular state trajectories are bounded by a decoupled, non-linear global integrity validator .

Hypothesis 1.2 (Adversarial Eigengap Robustness)**.** Adversarial injections targeting continuous online representation updates manifest geometrically as localized topological collapses within the empirical data covariance matrix . These collapses compress the structural eigengap ($\delta_k = \lambda_k - \lambda_{k+1} \to 0$) and induce rapid subspace divergence unless actively mitigated by dynamic noise-adaptive step-size contraction .

Definition 1.3 (Explicit Euler-Lagrange Guided Subspace)**.** The optimal, drift-resistant projection subspace matrix $W^* \in \mathbb{R}^{d\times k}$ is derived by solving the augmented regularized Euler-Lagrange action functional over the active data covariance operator C:

$$(C - \alpha GG^\top)W^* = W^*\Lambda \tag{1.1}$$

where $C = \mathbb{E}[xx^\top]$, $G \in \mathbb{R}^{d\times m}$ defines the real-time tracking constraint matrix, $\alpha > 0$ is the regularization penalty, and Λ is the diagonal matrix of augmented system eigenvalues .

Definition 1.4 (Explicit Obstacle Event)**.** An instantaneous structural change-point occurs at epoch t if and only if the intrinsic drift statistic G_t violates a localized threshold parameter τ :

$$\text{Event}_t \iff \|G_t\|_F > \tau \tag{1.2}$$

When triggered, the system initiates a flight-mode subspace realignment .

Definition 1.5 (Global Integrity Monitor (GIM) Predicates)**.** The multi-domain operational safety profile is governed by a real-time composite boolean predicate $I(t)$:

$$I(t) = I_{neural}(t) \wedge I_{spectral}(t) \wedge I_{symbolic}(t) \tag{1.3}$$

Stable, non-divergent execution is guaranteed if and only if $I(t) = 1$, strictly enforced by discrete hardware-level rollbacks prior to mathematical singularity convergence .

Definition 1.6 (MLRP on an Interval)**.** If the global density ratio $f(x)/g(x)$ fails to be globally monotonic, the domain $\mathcal{X}$ is partitioned into a finite set of locally monotone sub-intervals I_m such that the derivative of the ratio maintains a constant sign within each I_m, preserving threshold-optimal routing .

2 Primary Operational Theorems & Propositions

Theorem 2.1 (Constant-Complete GAS-ECF Margin Gain)**.** *For distinct prototypes c_i, c_j, projecting onto the guided action subspace W^* yields a strict lower-bound increase γ_{margin} in geometric separation relative to unconstrained PCA, for sufficiently small α :*

$$\|(W^*)^\top c_i - (W^*)^\top c_j\|_2^2 \geq \|W_{PCA}^\top c_i - W_{PCA}^\top c_j\|_2^2 + \gamma_{margin}(\alpha, G) \tag{2.1}$$

Theorem 2.2 (Constant-Explicit Event-Driven Subspace Tracking)**.** *The Quicksand Oja++ tracking updates uniformly converge to W^* . Between events, the asymptotic tracking error bound satisfies:*

$$\mathbb{E}[\|\hat{W}_t - W^*\|_F] \leq \mathcal{O}(\max\{\eta_t, \delta_t\}) \tag{2.2}$$

where η_t is the adaptive step-size and δ_t represents the hardware quantization noise variance .

Proposition 2.3 (Constant-Explicit Greedy Guarantee)**.** *The decentralized, step-wise memory selection sequence S executed over a submodular utility $f(S)$ satisfies the classical $(1 - 1/e)$ approximation bound, offset by the mathematically bounded distortion $\epsilon_{distortion}$ of the Spectral-Floor Patching operator :*

$$f(S) \geq \left(1 - \frac{1}{e}\right) \max_{|S^*| \leq K} f(S^*) - \epsilon_{distortion}(\tilde{K}_\alpha) \tag{2.3}$$

Theorem 2.4 (Non-Circular GIM-Induced Contraction)**.** *The parameter state trajectory mapping under the discrete symbolic domain is non-divergent . For a bounded perturbation $\Delta\theta_{neural}$:*

$$\|\mathcal{M}(\theta_t + \Delta\theta_{neural}) - \mathcal{M}(\theta_t)\|_{symbolic} \leq L_{contract}\|\Delta\theta_{neural}\|_2 \tag{2.4}$$

where the Lipschitz module $L_{contract} = |1 - 2\eta_t^{eff}\delta_k| + \mathcal{O}(\eta_t^2) < 1$ is actively maintained by the bounded operational step-size, completely removing the circular assumption of axiomatic contraction.

Theorem 2.5 (Explicit Hierarchical Error Bound)**.** *The total operational failure probability P_{top} across L stacked processing layers is bounded multiplicatively by independent conditional failure rates p_l and the stochastic hardware noise floor ϵ_{misc} :*

$$P_{top} \leq \sum_{l=1}^{L} p_l + \epsilon_{misc} \tag{2.5}$$

Theorem 2.6 (Stochastically-Bounded Hardware Realizability). *A modular processing substrate realizes the PMM state graph with high probability if its native computational primitives fulfill the lower bounds for parallel throughput and bandwidth relative to a stochastic latency distribution .*

Theorem 2.7 (Hierarchical Memory Compression). *The continuous iterative sequence of prototype updates $\{\tilde{x}_t\}$ converges asymptotically to a closed invariant subset . The footprint expands at a strictly sub-linear rate: $\mathcal{O}(\log t)$.*

Theorem 2.8 (Reversible Self-Modification). *A parameter update u_t is committed ($\theta_{t+1} = \theta_t + u_t$) if and only if $V_{global} \wedge V_{local}(u_t) = 1$. Otherwise, an automated digital SRAM snapshot rollback drives $\theta_{t+1} \leftarrow \theta_t$ with absolute zero residual precision drift .*

Theorem 2.9 (Dictionary Completeness). *Any discrete symbolic pattern exhibiting an infinite frequent occurrence signature within the continuous data stream will deterministically receive a permanent address in the dictionary D_t in finite time .*

Theorem 2.10 (Global-to-Local Validity). *An update preserves global stability and local logical soundness if and only if both the global validation operator V_{global} and local subsystem verifier V_{local} simultaneously return true .*

Theorem 2.11 (Memory Reconstruction). *Corrupted relational edges E_{ij} within the consolidated prototype memory graph can be reconstructed via exact nearest-neighbor matching using canonical representations $(\tilde{x}, D_t)$ with a bounded accuracy $\geq 95\%$.*

Theorem 2.12 (Internal Verification Completeness). *All symbolic validation checks and syntactic constraint deductions are fully resolvable within the autonomous shadow-worker loops, requiring no external ground-truth feedback .*

Theorem 2.13 (Strict Contraction of Memory Refinement). *The refinement operator $\Phi(x, y) = w_x x + w_y y$ over the parameter kernel $[\epsilon_0, 1 - \epsilon_0]$ acts as a strict contraction mapping under the Euclidean metric :*

$$\|\Phi(x_1, y) - \Phi(x_2, y)\|_2 \leq (1 - \epsilon_0)\|x_1 - x_2\|_2 \tag{2.6}$$

Theorem 2.14 (Robustness to Non-PSD Quantization Noise). *Submodular optimization guarantees are structurally preserved under the injection of non-PSD hardware quantization noise $\mathcal{E}_{noise}$, provided the operator norm satisfies $\|\mathcal{E}_{noise}\| \leq \frac{1}{2}(\lambda_k - \lambda_{k+1})$.*

Theorem 2.15 (Intrinsic Detector Concentration). *The intrinsic spectral-residual change-point detector G_t, derived over Riemannian curvature metrics, concentrates tightly around its mean. By Weyl's bounds, the transient realignment variance satisfies :*

$$P(\|G_t - \mathbb{E}[G_t]\| > \epsilon) \leq 2\exp\left(-\frac{2\epsilon^2 t}{\sigma^2_{manifold}}\right) \tag{2.7}$$

Theorem 2.16 (Analytic Boundary Patch for SCEF). *Dynamically constructing an analytic Tikhonov-regularized boundary patch operator structurally shifts the spectrum away from the zero boundary to $[\delta, 1 + \delta]$, restoring an exponential uniform convergence rate of $\mathcal{O}(e^{-M\chi})$ for the Stochastic Chebyshev Expansion Filter .*

Theorem 2.17 (Global iISS Loop Stability). *The interconnected feedback loop between the continuous neural domains and discrete logic cores is globally Integral Input-to-State Stable (iISS) . Under continuous-time idealization on the Stiefel manifold, the Lyapunov dissipation is governed tightly by the derived Euclidean coefficient :*

$$\dot{V}(U_t) \leq -2\delta_k V(U_t) \tag{2.8}$$

Theorem 2.18 (Geometric Distortion Limits for LSFP). *Applying the Localized Spectral-Floor Patching (LSFP) bounds the total induced geometric distortion strictly under the Frobenius norm :*

$$\|\tilde{K}_\alpha - K_\alpha\|_F \leq \delta\sqrt{M_0} \tag{2.9}$$

Theorem 2.19 (Lyapunov Stability for Quicksand Oja++). *The online event-driven subspace updates are globally stabilized. The difference operator of the continuously differentiable Lyapunov scalar function $V(x_t)$ satisfies :*

$$\Delta V(x_t) \leq -\eta_t \psi(x_t) + \mathcal{O}(\max\{\eta_t^2, \sigma_{hw}^2\}) \tag{2.10}$$

forcing the tracking error back into a stable bounding ball .

Theorem 2.20 (CHS Electromagnetic Decoupling). *A monolithic 3D neuromorphic integration substrate stochastically attenuates parasitic mutual inductive coupling. By wrapping interconnects in a Graphene/h-BN coaxial sleeve, the magnetic vector potential $A_{magnetic}$ is bounded strictly by the shield leakage variance ξ_{EM} .*

3 Supporting Lemmas & Spectral Bounds

Lemma 3.1 (Entropy Monotonicity). *Under stable long-term consolidation regimes, the spectral von Neumann entropy $\hat{H}_{vN}(t)$ computed over the canonical prototype covariance density matrix is strictly monotonic and non-increasing .*

Lemma 3.2 (GIM Mandatory Necessity). *Any hybrid architecture utilizing non-isomorphic internal state space manifolds ($\mathcal{M}_1 \ncong \mathcal{M}_2$) mathematically requires a globally decoupled validation monitor to prevent tracking collapse and coordinate decoupling .*

Lemma 3.3 (Spectral Drift Suppression Bound). *The real-time Oja++ stochastic approximation updates are deterministically bounded between change-point events according to $dist(\hat{W}_{t+1}, \hat{W}_t) \leq \eta_t \|x_t x_t^\top\|_F \leq \gamma_{suppression}$.*

Lemma 3.4 (Frequency-Type Partitioning Rule). *The Type-Conditioned Spectral Sorter (TCS) splits high-dimensional continuous activations into mutually orthogonal frequency bands: $\langle x^{(i)}, x^{(j)} \rangle = 0$ for all $i \neq j$.*

Lemma 3.5 (Cognitive Staircase Transform Principle). *The multi-level projection operator $\mathcal{T}_{staircase}$ maps volatile, high-entropy neural activation states into stable, low-entropy symbolic logic via quantized eigenvalue filtration .*

Lemma 3.6 (Piecewise-MLRP Threshold). *Segmenting a non-monotonic global density ratio into locally monotone sub-intervals allows the optimal cognitive routing policy to collapse at each boundary to a strict, single-threshold decision rule .*

Lemma 3.7 (Submodularity for Mixed Utilities). *A joint utility function defined as $f(S) = \log\det(I + \sigma^{-2}K_S) + \beta G(\mathcal{M}_S)$ is strictly submodular for all combinations of continuous prototypes and discrete logical atoms, provided $\beta \geq 0$.*

Lemma 3.8 (Intrinsic Spectral Drift Detector Stability). *By decoupling empirical and population spectra via Weyl's theorem, tracking variance concentrates tightly: $Var(G_t) \leq \frac{c_1}{t(\lambda_k - \lambda_{k+1})^2}$, minimizing false-alarm triggers .*

4 Structural Mathematical Invariants & Flow Constraints

Invariant 4.1. Memory Equivalence Relation: An immutable algebraic equivalence relation is enforced over incoming elements: $x \sim y \iff \text{Type}(x) \equiv \text{Type}(y) \wedge \|x - y\|_2 \leq \epsilon_{redundancy}$.

Invariant 4.2. Canonical Representative Structural Invariant: Every active equivalence class maps uniquely to a single, condensed canonical prototype vector .

Invariant 4.3. Hierarchical Refinement Convex Invariant: Parametric updates satisfy convex hull properties: $w_x + w_y = 1$ and $w_x, w_y \geq 0$.

Invariant 4.4. Monotone Contractive Invariant: The consolidation operator preserves the contractive magnitude bound $\|\Phi(x, y)\|_2 \leq \max(\|x\|_2, \|y\|_2)$.

Invariant 4.5. Parameter Trace Tuple Invariant: State mutations are tracked via an ordered tuple $T_t = (\theta_t, u_t)$.

Invariant 4.6. Tri-Domain Transparency Invariant: Data streams traversing continuous-to-discrete interfaces must preserve underlying semantic weights under inverse kernel operations .

Invariant 4.7. Deterministic-Probabilistic Compatibility: The state transition matrix is valid if and only if the system variances are jointly authorized by the GIM matrix $\mathcal{C}_{gim} \equiv 1$.

Invariant 4.8. Asynchronous Safety Invariant: Background parallel paths commit parameters if and only if updates do not drive the active global GIM error above zero .

Invariant 4.9. Three-Timescale Training Clock: Parameters are decoupled to eliminate execution hazards: $\Delta T_{neural} \ll \Delta T_{spectral} \ll \Delta T_{symbolic}$.

Invariant 4.10. Monotone Confidence Rule: Local verification confidence metrics $\mathcal{C}_{local}$ monotonically increase or plateau across sequential deductions .

Invariant 4.11. Multi-Engine Synchronization: Continuous parameter adjustments must trigger a proportional shift in the energy landscape of the Symbolic Microcode Core .

Invariant 4.12. Safety Temporal Logic (STL) Contract: Global integrity violations force an instantaneous hardware state stabilization within a bounded time window $\tau_{rollback}$.

5 Hardware-Aware Stochastic Boundary Conditions

Absolute microarchitectural physical targets are herein formalized as bounds operating under a defined environmental noise distribution $\xi_{hw} \sim \mathcal{N}(0, \Sigma_{hw})$.

Stochastic Boundary Condition 5.1 (Combinational Latency)**.** The routing network resolves sparse pointer structures bounded by the stochastic propagation limit $\tau_{latency} \leq \tau_{gate} \cdot \lceil \log_2 k \rceil \in \mathcal{O}(1)$.

Stochastic Boundary Condition 5.2 (Stochastic Thermal Dissipation)**.** Temperature variations ΔT driven by processing volumes are structurally dissipated by vertical CNT pillars. The arithmetic precision degradation $\epsilon_{arith}(\xi_{thermal})$ remains stochastically bounded below 10^{-3} .

Stochastic Boundary Condition 5.3 (Interfacial Mechanical Shear)**.** The intermediate graded $Si_{1-x}Ge_x$ buffer layer dynamically absorbs thermal shockwaves such that the probability of the peak transient stress exceeding the yield point approaches zero .

Stochastic Boundary Condition 5.4 (Hysteresis Dead-Zone Rank)**.** The active tracking rank k_t evolves via a step-wise hysteresis dead-zone controller to filter ambient thermal noise and prevent rapid structural oscillation .

Stochastic Boundary Condition 5.5 (Localized Spectral-Floor Patching Matrix)**.** The synthesis of the patched Gram matrix $\tilde{K}_\alpha$ utilizes the precision stability floor δ to bound quantization entropy .

Stochastic Boundary Condition 5.6 (Electromagnetic Shield Leakage)**.** The transient SOT-MRAM write noise is attenuated by the coaxial boundaries, bounding the inductive coupling scalar to a nominal leakage distribution $M_{link} = \delta_{shield} \cdot M_{raw}$.

Stochastic Boundary Condition 5.7 (DASM Rollback State Validation)**.** Following a GIM failure event F_ℓ, parameter state restoration is executed using parallel hardware DASM on digital SRAM, guaranteeing zero analog drift variance .

6 Formal Evaluation Test Suite & Verification Protocols

Protocol 6.1. Canonical Convergence: Verify strict asymptotic convergence of continuous memory vectors towards a stable prototype attractor .

Protocol 6.2. Rollback Safety: Forcing logic violations inside the loop must trigger single-cycle parameter restoration ($\theta_{t+1} \equiv \theta_t$) .

Protocol 6.3. Dictionary Growth: Ensure recurring structural patterns result in deterministic symbolic insertion .

Protocol 6.4. Global Stability: Adversarial perturbations must yield overall drift trajectories strictly bounded below the system threshold τ .

Protocol 6.5. Reconstruction Accuracy: Random edge deletion from the memory graph must satisfy recovery performance $\geq 95\%$ under nearest-neighbor mapping .

Protocol 6.6. Reconstruction Internal Verification: Validate bit-true matching between internal shadow-worker verifications and external ground-truth keys .

Protocol 6.7. GAS-ECF Margin Gain Validation: Execute paired t-tests over derived projections versus baseline PCA to mathematically reject equivalent separation margins .

Protocol 6.8. Quicksand Oja++ Tracking: Verify matrix Bernstein inequalities actively bound false-alarm rates during subspace drift phases .

Protocol 6.9. Tomographic PCA Recovery: Phase transitions governing exact subspace recovery must be verified with random quadratic probing $N_{probes} \geq \mathcal{O}(k \cdot d \log d)$.

Protocol 6.10. Discretizer Contraction: Verify noise-adaptive filters force continuous perturbations to compress deterministically into discrete assignments .

Protocol 6.11. End-to-End Error Bound: Verify the cumulative system failure cascade strictly obeys the theoretical multiplicative upper bound $\sum p_\ell + p_{src} + \epsilon_{arith}$ under maximum adversarial noise .